Project Proposal

Project Proposal: Food Truck Location Strategy and Profit Optimization

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Course: ISM6404 - Data Analytics

Dataset: Food Truck Daily Sales Dataset (March-August 2025)

Selected Dataset:

Our team selected the Food Truck Daily Sales Dataset, which contains daily operational and sales information for a mobile food truck operating across multiple locations from March through August 2025. The dataset includes more than 180 daily observations and contains both operational and environmental variables such as date, day of week, location, weather conditions, temperature, special events, number of competitors, customer count, items sold, revenue, daily cost, and profit. This dataset captures the dynamic environment in which food trucks operate. Demand and profitability are influenced by both internal decisions, such as location selection, and external factors including weather conditions, local events, and the presence of competing food trucks. Because these variables vary daily, the dataset provides an opportunity to analyze how environmental and operational factors interact to influence customer turnout and financial performance.

Problem Statement:

Tasty Wheels must decide where to operate each day to maximize profitability. However, food truck operations are highly sensitive to external conditions such as weather, temperature, special events, and competition. The central business problem of this project is to identify the key factors that drive daily customer turnout and profitability and use those insights to guide better operational decisions. Specifically, this project seeks to answer the following questions:

- Which environmental and operational factors strongly influence customer turnout?
- How do weather conditions, temperature, location, and special events affect revenue and profit?
- Does competitor presence significantly impact customer demand?
- Are there identifiable patterns in daily operations that correspond to higher profitability?

This problem is important because food trucks operate in highly variable environments. Understanding demand patterns enables better decisions about staffing, inventory, pricing, promotions, and location scheduling. By identifying these drivers, the project aims to help the food truck owner make data-driven decisions regarding location selection, event participation, staffing, and inventory planning.

Problem Type:

This project will involve multiple analytical components:

- **Regression:** Predict daily customers using temperature, weather, location, events, and competitors. Predict daily profit using customers, items sold, and environmental variables.
- **Clustering Analysis:** Clustering techniques will be used to identify patterns in operational conditions and group similar days together. This analysis will help identify common operational scenarios, such as high-demand event days, weather-sensitive days, or low-demand competitive environments. These clusters may reveal patterns that are not immediately visible through regression analysis alone and can support strategic planning and scheduling decisions.

Proposed Solution Methods:

We plan to use a combination of statistical and machine learning techniques:

Exploratory Data Analysis:

- Summary statistics of key variables
- Visualization of revenue, profit, and customer trends over time
- Boxplots and grouped comparisons across locations, weather conditions, and event types
- Analysis of weekday versus weekend demand
- Scatterplots exploring relationships between temperature, customers, and profit

Correlation and Multi-Collinearity Analysis:

- Correlation matrices to identify relationships among numerical variables
- Variance Inflation Factor (VIF) tests to assess potential multicollinearity among predictors
- Removal or adjustment of highly correlated variables if necessary

Simple Linear Regression:

- Explore direct relationships between variables, such as temperature vs. customers or customers vs. profit

Multiple Linear Regression:

- Model profit as a function of environmental and operational variables including temperature, weather, location, events, and competitors

Cross-Validation:

- Evaluate model stability and reduce the risk of overfitting.

Model-Evaluation:

- To evaluate model performance, we will use several methods depending on model type to ensure the selected models provide reliable and interpretable results.
    - R2 and Adjusted R2 will be used to measure explanatory power of regression models.
    - Root Mean Squared Error will be used to evaluate prediction accuracy.

Clustering Analysis:

- To identify patterns across operating conditions, we will apply k-means clustering to group similar days based on variables such as temperature, customers, revenue, and profit. Clustering results will help identify operational scenarios such as:
    - High-demand event days
    - Weather-sensitive days
    - Low-demand competitive environments
    - Consistently profitable operating conditions

Expected Outcomes:

The expected outcomes of this project include both analytical insights and practical recommendations:

Analytical Insights:

- Identification of the strongest predictors of customer turnout
- Quantification of the impact of weather and temperature on demand
- Measurement of how special events influence revenue and profit
- Evaluation of the effect of competitor presence on daily performance
- Identification of operational patterns through clustering analysis

Predictive Models:

- A model that forecasts daily customers with reasonable accuracy.
- A model that predicts daily profit to support planning and resource allocation.

Business Recommendations:

- Based on these findings, the project will provide recommendations to support data-driven decision making. These recommendations may include:
    - Prioritizing high-performing locations on specific days of the week
    - Increasing staffing and inventory during high-demand event days
    - Adjusting operations during adverse weather conditions
    - Using competitor presence to inform location selection strategies
    - Improving weekly route planning based on demand patterns

Operational Improvements:

- These insights can ultimately help improve operational efficiency through:
    - More accurate inventory planning
    - Better staffing schedules
    - More strategic event participation
    - Improved daily location selection